



# Design Guide for Abu Dhabi

Minimum design standards for public health services





# **Design Guide for Abu Dhabi:**

Minimum design standards for public health services

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This document aims at establishing the minimum design standards for the design of public health systems for buildings in Abu Dhabi. It does not relieve the designer of responsibility for accurately determining design arrangements and/or complying with the local authority requirements and/or relevant international codes or regulatory requirements. It should be used for initial design guidance, and provides minimum design requirements for specific design issues.

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# 1 Design codes, standards and guidance

The design, specification and installation of public health services should be in accordance with good engineering practice and should conform to, but not be limited to, the following standards, codes of practice and regulations applicable to Abu Dhabi:

- Abu Dhabi International Building Code (ADIBC) regulations
- Abu Dhabi Regulations and Supervisory Bureau regulations
- Abu Dhabi Department of Municipal Affairs and Transport regulations
- Abu Dhabi Food Control Authority regulations
- Abu Dhabi Environment Agency regulations
- Abu Dhabi Distribution Company (ADDC) regulations
- Abu Dhabi Municipality (ADM) regulations
- Abu Dhabi Sewerage Services (ADSSC) regulations
- *UAE Fire and Life Safety Code of Practice* (ADCD, 2018)
- Abu Dhabi Urban Planning Council
- Abu Dhabi ADNOC Distribution Guidelines
- Uniform Plumbing Code of Abu Dhabi (UPC)
- National Fire Protection Association (NFPA)
- International Plumbing Code (IPC)
- American Society of Plumbing Engineers (ASPE)
- British Standards (BSI)
- European Norms (EN)
- International Organisation for Standardisation standards (ISO)
- Water Research Centre publications
- The Chartered Institute of Plumbing and Heating Engineering (CIPHE) good practice and design guides.

The public health systems covered in this section are as follows:

- local green building requirements
- domestic water systems (hot and cold)
- drainage systems
- rainwater systems
- gas systems.

The main objectives should be to:

- ensure that all water supplies are clear, odourless, tasteless and wholesome for use
- ensure that the system is protected against *Legionella*
- provide potable water in the amount and at the pressure required by the building occupancy and type of plumbing fixtures utilised in the building
- prevent contamination from non-potable liquids, solids or gases
- prevent water hammer conditions by providing air chambers and correct bracketry
- minimising leakage and wastage.

## 2 Local green building requirements

Because of Abu Dhabi's limited annual rainfall, hot climate, and the significant energy embodied in potable water due to desalinization, water conservation is a priority for Estidama. As discussed in the April 2006 UAE Initial National Communication to the United Nations on climate change, it is a distinct possibility that the UAE will become even drier due to the effects of global warming. Coupled with projected population increases, the importance of accelerating water conservation and reuse efforts is clear.

'One of the most important challenges for the Emirate is to balance water supply and demand as efficiently as possible given that the per capita consumption of fresh water is among the highest in the world and new water supplies are expensive' — Abu Dhabi Water Resources Master Plan, EAD 2009

Within Estidama, the 'Precious Water' section of the *Pearl Building Rating System: Design and Construction* (ADUPC, 2010) includes the Building Water Calculator that should be used throughout the entire design process. This calculator will help the development team to assess inputs and flows of water to and from the project with a holistic perspective, and assist in identifying how and where reductions in the use of potable water may be made.

The following table defines the water services maximum flow rates based on: Estidama Precious Water Credit PW-R1: Minimum interior water use reduction.

**Table 2.1** Maximum flow rates for water services

| Fixture of fixture fitting | Percentage reduction                 |
|----------------------------|--------------------------------------|
| Bathroom taps, private     | 6 litres/min. at 417.7 kPa           |
| Bathroom taps, public      | 1.9 litres/min. at 417.7 kPa         |
| Shower head                | 9.5 litres/min. at 551.6 kPa         |
| Kitchen sink faucet        | 6 litres/min. at 417.7 kPa           |
| Bidets                     | 6 litres/min.                        |
| Urinal                     | 0.5 litres/flushing cycle            |
| Toilets (dual flush)       | 6/4 litres/flushing cycle (full/low) |

## 3 Domestic cold water services, design criteria, codes, standards and regulations

The design, specification and installation of cold water systems should be in accordance with good engineering practice and should conform to, but not be limited to, the following standards, codes of practice and regulations applicable to Abu Dhabi:

- RSB regulations
- ADDC regulations
- ADM requirements
- CIBSE Guide G: *Public health and plumbing engineering* (CIBSE, 2014)
- BS EN 806 (BSI, 2000–2010, 2015)
- *Plumbing Engineering Services Design Guide* (IoP, 2002)

Maintain a maximum velocity in the water supply piping as per the following table, from Table 17 on page 16 of the *Plumbing Engineering Services Design Guide* (IoP, 2002), to prevent noise and decrease the danger of surge pressure shock.

**Table 3.1** Maximum velocities for water supply pipes

| Location                               | Noise rating (NR) | Metal pipes (m/s) | Plastic pipes (m/s) |
|----------------------------------------|-------------------|-------------------|---------------------|
| Service duct, riser, shaft, plant room | 50                | 2.0               | 2.5                 |
| Service enclosure, ceiling void        | 40                | 1.5               | 1.5                 |
| Circulation area, entrance corridor    | 35                | 1.5               | 1.5                 |
| Seating area, lecture/meeting room     | 30                | 1.25              | 1.25                |
| Bedroom                                | 25                | 1.0               | 1.0                 |
| Theatre, cinema                        | 20                | 0.75              | 0.75                |
| Recording studios                      | Less than 20      | 0.5               | 0.5                 |

## 4 Water storage design criteria

The following storage figures have been taken from Table 1.1 in Issue 3 of the *Guide to Water Supply Regulations* (ADRSB, 2017) to calculate the one-day total storage requirement. The *Guide to Water Supply Regulations* recommends that a total of one to two days storage should be allowed for in all buildings, therefore two times the daily volume should form the basis of the total on-site water storage.

The water supply demand and storage requirements should be validated or approved by ADDC in the early project stage.

**Table 4.1** Estimates of daily water consumption

| Type of premises and consumption category | Description              | Estimates of daily rate of consumption (Imperial gallons) | Rounded estimates of daily rate of consumption (litres) |
|-------------------------------------------|--------------------------|-----------------------------------------------------------|---------------------------------------------------------|
| Hotel <sup>[1]</sup>                      | Per one bed              | 100–150                                                   | 450–675                                                 |
| Car wash station (CWS)                    | Per manual channel       | 800                                                       | 3600                                                    |
|                                           | Per automatic channel    | 2400                                                      | 10800                                                   |
| Hospital                                  | Per one bed              | 150                                                       | 675                                                     |
| Day clinic                                | Per medical practitioner | 100                                                       | 450                                                     |
| Day clinic (with dental)                  | Per medical practitioner | 150                                                       | 675                                                     |
| Common market                             | Per square metre         | 1.1                                                       | 5                                                       |
| Mosques < 300 m <sup>2</sup>              | Per square metre         | 5.5                                                       | 25                                                      |
| Mosques > 300 m <sup>2</sup>              | Per square metre         | 3.5                                                       | 16                                                      |
| Female praying rooms                      | Per square metre         | 2                                                         | 9                                                       |
| Workers' housing                          | Per capita               | 44                                                        | 200                                                     |
| Public toilets                            | Per sanitary piece       | 35–50                                                     | 160–225                                                 |
| Schools/universities                      | Per student              | 5/10                                                      | 25/45                                                   |
| Hostel                                    | Per student              | 44                                                        | 200                                                     |
| Villa and shabiat                         | Per capita               | 77                                                        | 350                                                     |
| Villa/shabiat                             | Per small service block  | 250                                                       | 1100                                                    |
|                                           | Per large service block  | 450                                                       | 2000                                                    |
|                                           | Per external majilis     | 150                                                       | 675                                                     |
|                                           | Per maid's room          | 50                                                        | 225                                                     |
|                                           | Per guard room           | 100                                                       | 450                                                     |

Table continues

**Table 4.1** Estimates of daily water consumption (*continued*)

| Type of premises and consumption category | Description                         | Estimates of daily rate of consumption (Imperial gallons) | Rounded estimates of daily rate of consumption (litres) |
|-------------------------------------------|-------------------------------------|-----------------------------------------------------------|---------------------------------------------------------|
| Villa and shabiat                         | Per bedroom <sup>[2]</sup>          | 110                                                       | 500                                                     |
|                                           | Per square metre swimming pool area | 4–5                                                       | 18–22                                                   |
| General services <sup>[3]</sup>           | Per plot square metre               | 0.2                                                       | 0.9                                                     |
| Services in a building                    | Up to 5 floors                      | 165                                                       | 750                                                     |
|                                           | From 6 to 10 floors                 | 330                                                       | 1500                                                    |
|                                           | Above 10 floors <sup>[4]</sup>      | 660                                                       | 3000                                                    |
| Offices and shops                         | Per sanitary piece <sup>[5]</sup>   | 35–50                                                     | 160–225                                                 |
|                                           | Per square metre                    | 1                                                         | 4.5                                                     |
|                                           | Per person                          | 10                                                        | 9                                                       |
| Restaurants                               | Per meal                            | 2                                                         | 45                                                      |
| Residential flat                          | Studio                              | 100                                                       | 450                                                     |
|                                           | 1 bedroom                           | 120                                                       | 550                                                     |
|                                           | 2 bedrooms                          | 180                                                       | 820                                                     |
|                                           | 3 bedrooms                          | 220                                                       | 1000                                                    |
|                                           | 4 bedrooms                          | 280                                                       | 1250                                                    |
|                                           | 5 bedrooms                          | 350                                                       | 1600                                                    |
|                                           | Per maid's room                     | 50                                                        | 225                                                     |
|                                           | Per capita                          | 50                                                        | 225                                                     |

Table notes:

- 1 Hotel category up to five-star; hotels/resorts above five stars will be subject to assessment.
- 2 For the shabiat and villa category, a reduction factor may be applied for every additional bedroom according to the Distribution Company's own criteria.
- 3 'General services' means water used for internal gardening and general cleaning purposes for a standard-size shabiat and villa.
- 4 Rates of consumption for buildings higher than 20 floors shall be adjusted proportionally.
- 5 All consumption rates for sanitary pieces shall be calculated based on water-efficient plumbing fittings, as required by Estdama.

Note (a): Water consumption rates for other uses, e.g. swimming pools, cooling services, shall be determined following consultation with the Distribution Company.

Note (b): For some of the categories, the Responsible Person has a choice of consumption rate which is subject to the Distribution Company approval.

## 5 Pipe sizing

The size of water service mains, branch mains and risers should be determined according to water supply demand (litre/min.), available water pressure (kPa) and friction loss due to the water meter and developed length of pipe (m), including equivalent length of fittings.

Supply load in the building water distribution system shall be determined by total load on the piped being sized, in terms of water supply fixture units (WSFU), as shown in Table 6-7, 'Separate Cold and Hot Water Supply Fixture Unit Values' in the *Uniform Plumbing Code of Abu Dhabi* (ADEA, 2009).

**Table 5.1** Water supply fixture units

| Appliances, appurtenances or fixtures <sup>[1]</sup>   | Minimum fixture branch pipe size (mm) <sup>[2]</sup> | Occupancy                   | Water supply fixture units (WSFU) |       |       |
|--------------------------------------------------------|------------------------------------------------------|-----------------------------|-----------------------------------|-------|-------|
|                                                        |                                                      |                             | Cold                              | Hot   | Total |
| Bathtub or bath/shower                                 | 15                                                   | Private                     | 1.0                               | 1.0   | 1.4   |
|                                                        |                                                      | Public                      | 3.0                               | 3.0   | 4.0   |
| Bidet                                                  | 15                                                   | Private                     | 1.0                               | —     | 1.0   |
| Clothes washer                                         | 15                                                   | Private or public           | 3.0                               | 3.0   | 4.0   |
| Dental Unit, cuspidor                                  | 15                                                   | Public                      | 1.0                               | —     | 1.0   |
| Dishwasher, domestic                                   | 15                                                   | Private or public           |                                   | 1.5   | 1.5   |
| Drinking fountain                                      | 15                                                   | Private or public           | 0.5                               | —     | 0.5   |
|                                                        |                                                      | Assembly                    | 0.75                              | —     | 0.75  |
| Hose bibb                                              | 15                                                   | Private or public           | 2.5                               | —     | 2.5   |
| Hose bibb, each additional when used with total demand | 15                                                   | Private or public           | 1.0                               | —     | 1.0   |
| Lavatory                                               | 15                                                   | Private, public or assembly | —                                 | —     | —     |
| Lawn sprinkler, each head <sup>[3]</sup>               |                                                      | Private or public           | 5.0                               | —     | 5.0   |
| Sinks:                                                 |                                                      |                             |                                   |       |       |
| — bar                                                  | 15                                                   | Private                     | 0.75                              | 0.75  | 1.0   |
|                                                        |                                                      | Public                      | 1.5                               | 1.5   | 2.0   |
| — clinic (faucet)                                      | 15                                                   | Public                      | 2.25                              | 2.25  | 3.0   |
| — clinic (flushometer valve with or without faucet)    | 25                                                   | Public                      | 8.0                               | —     | 8.0   |
| — kitchen, domestic                                    | 15                                                   | Private or public           | 1.125                             | 1.125 | 1.5   |
| — laundry                                              |                                                      | Private or public           | 1.125                             | 1.125 | 1.5   |
| — service or mop basin                                 | 15                                                   | Private                     | 1.125                             | 1.125 | 1.5   |
|                                                        |                                                      | Public                      | 2.25                              | 2.25  | 3.0   |
| — washup, each set of faucets                          | 15                                                   | Public                      | 1.5                               | 1.5   | 2.0   |
| Shower, per head                                       | 15                                                   | Private or public           | 1.5                               | 1.5   | 2.0   |
| Urinal (2 litre/flush flushometer valve)               | 20                                                   | Public                      | 4.0                               | —     | 4.0   |
|                                                        |                                                      | Assembly                    | 5.0                               | —     | 5.0   |
| Urinal, flush tank                                     | 15                                                   | Private or public           | 2.0                               | —     | 2.0   |
|                                                        |                                                      | Assembly                    | 3.0                               | —     | 3.0   |
| Wash fountain, circular spray                          | 20                                                   | Public                      | 3.0                               | 3.0   | 4.0   |
| Water closet (6 litre/flush gravity tank)              | 15                                                   | Private or public           | 2.5                               | —     | 2.5   |
|                                                        |                                                      | Assembly                    | 3.5                               | —     | 3.5   |
| Water closet (6 litre/flush flushometer tank)          | 15                                                   | Private or public           | 2.5                               | —     | 2.5   |
|                                                        |                                                      | Assembly                    | 3.5                               | —     | 3.5   |
| Water closet (6 litre/flush flushometer valve)         | 25                                                   | Public                      | 5.0                               | —     | 5.0   |
|                                                        |                                                      | Assembly                    | 8.0                               | —     | 8.0   |

**Notes:**

- 1 Appliances, fixtures and appurtenances not included in this table may be sized by reference to fixtures having similar flow rates and frequency of use.
- 2 The listed minimum supply branch pipe sizes for individual fixtures are the nominal (ID) pipe size.
- 3 For fixtures or supply connections likely to impose continuous flow demands, determine the required flow in litres/min. and add separately to the demand in litres/min. for the distribution system of portions thereof.

Unit conversion: 1 mm = 0.04 in.; 1 litre/min = 0.26 gal./min.

To convert the WSFUs into flow rates, the corresponding value should be read from the *Uniform Plumbing Code* (ADEA, 2009) Table 6-9, 'Conversion of Water Supply Fixture Units Equivalent to L/min', provided below as Table 5.2.

**Table 5.2** Conversion of water supply fixture units to flow rates

| Demand (load)              |                                       |                                        | Demand (load)              |                                       |                                        |
|----------------------------|---------------------------------------|----------------------------------------|----------------------------|---------------------------------------|----------------------------------------|
| Water supply fixture units | System with flush tanks (litres/min.) | System with flush valves (litres/min.) | Water supply fixture units | System with flush tanks (litres/min.) | System with flush valves (litres/min.) |
| 1                          | —                                     | —                                      | 160                        | 216                                   | 314                                    |
| 2                          | 3.8                                   | —                                      | 180                        | 231                                   | 329                                    |
| 3                          | 11                                    | —                                      | 200                        | 246                                   | 348                                    |
| 4                          | 15                                    | —                                      | 225                        | 265                                   | 367                                    |
| 5                          | 19                                    | —                                      | 250                        | 284                                   | 382                                    |
| 6                          | 23                                    | —                                      | 275                        | 303                                   | 401                                    |
| 8                          | 25                                    | —                                      | 300                        | 322                                   | 416                                    |
| 10                         | 30                                    | 102                                    | 400                        | 397                                   | 477                                    |
| 12                         | 34                                    | 110                                    | 500                        | 473                                   | 537                                    |
| 14                         | 42                                    | 114                                    | 750                        | 643                                   | 674                                    |
| 16                         | 45                                    | 121                                    | 1000                       | 787                                   | 787                                    |
| 18                         | 49                                    | 125                                    | 1250                       | 908                                   | 908                                    |
| 20                         | 53                                    | 132                                    | 1500                       | 1011                                  | 1011                                   |
| 25                         | 64                                    | 144                                    | 1750                       | 1113                                  | 1113                                   |
| 30                         | 76                                    | 155                                    | 2000                       | 1215                                  | 1215                                   |
| 35                         | 87                                    | 167                                    | 2250                       | 1317                                  | 1317                                   |
| 40                         | 95                                    | 178                                    | 2500                       | 1419                                  | 1419                                   |
| 45                         | 102                                   | 185                                    | 2750                       | 1522                                  | 1522                                   |
| 50                         | 110                                   | 197                                    | 3000                       | 1635                                  | 1635                                   |
| 60                         | 121                                   | 208                                    | 4000                       | 1987                                  | 1987                                   |
| 70                         | 132                                   | 223                                    | 5000                       | 2245                                  | 2245                                   |
| 80                         | 144                                   | 235                                    | 6000                       | 2434                                  | 2434                                   |
| 90                         | 155                                   | 246                                    | 7000                       | 2593                                  | 2593                                   |
| 100                        | 167                                   | 257                                    | 8000                       | 2718                                  | 2718                                   |
| 120                        | 182                                   | 276                                    | 9000                       | 2820                                  | 2820                                   |
| 140                        | 201                                   | 295                                    | 10 000                     | 2911                                  | 2911                                   |

Unit conversion: 1 litre/min. = 0.26 gal./min.

To convert flow rates into pipe sizing, the corresponding flow rate values should be utilised with *Uniform Plumbing Code* pipe selection charts: Chart A-4, Chart A-5, Chart A-6 and Chart A-7. Where specific manufacturer's pipework has been specified the manufacturer's information can be utilised for pipe sizing.

## 6 Legionella control

In order to prevent the growth of Legionnaires' disease bacteria in warm water systems, the design installation and maintenance of these systems must be in accordance with the Abu Dhabi EHS Code of Practice 12.0: *Prevention and Control of Legionnaires' Disease* (ADOHSC, 2017) and ASHRAE Guidelines 12-2000: *Minimizing the Risk of Legionellosis Associated with Building Water Systems* (ASHRAE, 2020).

Bacterial protection should be incorporated to protect against the risk of *Legionella*; the acceptable primary forms of water treatment are as below:

- Temperature control: It is recommended that hot water should be stored at 60 °C and distributed so that, after one minute of running, a temperature of at least 50 °C is attainable at outlets. Cold water storage and distribution should be at 20 °C or below.
- Pasteurisation: Water systems must be flushed at least every six months with water at not less than 70 °C for 5 minutes (or an equivalent temperature/time combination) or treated with chlorine to provide minimum free chlorine of 1–2 mg/l at all outlets.
- Cu-Ag ionization treatment.
- Chlorination is the process of applying any of the chlorine-containing compounds such as sodium hypochlorite (bleach solution) to the water to achieve the necessary destruction of all bacteria. The maximum chlorine dosing shall not exceed 0.5 mg/l and the residual chlorine at any fixtures within the water distribution system shall be at least 0.2 ppm (mg/l) as per WHO standards.
- Chlorine dioxide is an oxidizing biocide capable of reacting with a wide range of organic substances. There are commercial systems available that release chlorine dioxide from a stabilized precursor solution into water systems. The maximum value of chlorine dioxide in water should not exceed 0.5 mg/l as chlorine dioxide. Chlorine dioxide level of 0.5 mg/l can, if properly managed, be effective against planktonic *Legionella* in both hot and cold water systems.
- Ultraviolet (UV) light and ozone (O<sub>3</sub>): The previous treatment systems are effective throughout the water system downstream to the point of application. UV and O<sub>3</sub> can be used to treat water at locations very close to the point of application.

The chlorination system should be designed carefully to ensure that there is no over-chlorination dosing to the system, as this could affect the life span of the pipe material.

## 7 Domestic hot water services, design criteria, codes, standards and regulations

The design, specification and installation of hot water systems should be in accordance with good engineering practice and should conform to, but not be limited to, the following standards, codes of practice and specifications:

- *Uniform Plumbing Code of Abu Dhabi Emirate* (ADEA, 2009)
- *Public health and plumbing engineering* (CIBSE, 2014)
- Chartered Institute of Plumbing and Heating Engineers (CIPHE) guidance
- American Society of Heating, Refrigeration and Air conditioning Engineers (ASHRAE) guidance
- Abu Dhabi Municipality requirements.

The following are design temperatures considered for the domestic hot water (DHW) system:

- Minimum DHWS storage temperature: 60 °C
- Minimum temperature at outlet: 55 °C
- Minimum temperature of return to storage cylinders: 50 °C.

The hot water temperature for appliances, where applicable, should be temperature-controlled via a thermostatic mixing valve (TMV) as detailed within Table 7.1 below, taken from the *Plumbing Engineering Services Design Guide* (IoP, 2002).

**Table 7.1** Water temperatures for thermostatic mixing valves  
(source: *Plumbing Engineering Services Design Guide* (IoP, 2002))

| Appliance       | Application | Thermostatic mixing valve |      |
|-----------------|-------------|---------------------------|------|
|                 |             | Temp. (°C)                | Type |
| Bidet           | All         | 38                        | 3    |
| Shower          | All         | 41                        | 3    |
| Wash hand basin | Public      | 41                        | 1    |
|                 | Private     | 43                        | 3    |
| Bath            | All         | 44                        | 3    |

The hot water storage figures in Table 7.2 have been taken from Table 6 of the *Plumbing Engineering Services Design Guide* (IoP, 2002) to assist in calculating daily hot water requirements and storage for a building, based on a two-hour reheat period, an inlet temperature of 10 °C and a stored temperature of 65 °C. The design engineer should make the necessary adjustments depending on the actual reheat period, inlet water temperature and storage temperature.

**Table 7.2** Recommended hot water storage temperatures (source: *Plumbing Engineering Services Design Guide* (IoP, 2002))

| Type of building                 | Daily consumption (litres) | Stored (litres) | Unit          |
|----------------------------------|----------------------------|-----------------|---------------|
| Dwellings:                       |                            |                 |               |
| — 1 bedroom                      | 115                        | 115             | Per bedroom   |
| — 2 bedrooms                     | 75                         | 115             | Per bedroom   |
| — 3+ bedrooms                    | 55                         | 115             | Per bedroom   |
| — student, en-suite              | 70                         | 20              | Per bedroom   |
| — student, communal              | 70                         | 20              | Per bed space |
| — nurses home                    | 70                         | 20              | Per bed space |
| — children's home                | 70                         | 25              | Per bed space |
| — elderly sheltered              | 70                         | 25              | Per bed space |
| — elderly care home              | 90                         | 25              | Per bed space |
| Hotels:                          |                            |                 |               |
| — budget                         | 115                        | 35              | Per bedroom   |
| — Travel Inn/Lodge               | 115                        | 35              | Per bedroom   |
| — luxury (4/5-star)              | 135                        | 45              | Per bedroom   |
| Offices and general work places: |                            |                 |               |
| — with canteen                   | 15                         | 5               | Per person    |
| — without canteen                | 10                         | 5               | Per person    |
| Shops:                           |                            |                 |               |
| — with canteen                   | 15                         | 5               | Per person    |
| — without canteen                | 10                         | 5               | Per person    |
| Factories:                       |                            |                 |               |
| — with canteen                   | 15                         | 5               | Per person    |
| — without canteen                | 10                         | 5               | Per person    |

*Table continues*

**Table 7.2** Recommended hot water storage temperatures (*continued*)  
(source: *Plumbing Engineering Services Design Guide* (IoP, 2002))

| Type of building                      | Daily consumption (litres) | Stored (litres) | Unit       |
|---------------------------------------|----------------------------|-----------------|------------|
| Schools:                              |                            |                 |            |
| — nursery                             | 15                         | 5               | Per pupil  |
| — primary                             | 15                         | 5               | Per pupil  |
| — secondary                           | 15                         | 5               | Per pupil  |
| — 6th form                            | 15                         | 5               | Per pupil  |
| — boarding                            | 114                        | 25              | Per pupil  |
| Hospitals:                            |                            |                 |            |
| — district general                    | 200                        | 50              | Per bed    |
| — surgical ward                       | 110                        | 50              | Per bed    |
| — medical ward                        | 110                        | 50              | Per bed    |
| — paediatric ward                     | 125                        | 70              | Per bed    |
| — geriatric ward                      | 70                         | 40              | Per bed    |
| Sports changing:                      |                            |                 |            |
| — sports hall                         | 20                         | 20              | Per person |
| — swimming pool                       | 20                         | 20              | Per person |
| — field sports                        | 35                         | 35              | Per person |
| — all-weather pitch                   | 35                         | 35              | Per person |
| Places of assembly (excluding staff): |                            |                 |            |
| — art gallery                         | 2                          | 1               | Per person |
| — library                             | 2                          | 1               | Per person |
| — museum                              | 1                          | 1               | Per person |
| — theatre                             | 1                          | 1               | Per person |
| — cinema                              | 1                          | 1               | Per person |
| — bar                                 | 2                          | 1               | Per person |
| — night club                          | 1                          | 1               | Per person |
| — restaurant                          | 6                          | 6               | Per cover  |

## 8 Foul and waste drainage systems: design criteria, codes, standards and regulations

The design, specification and installation of drainage systems should be in accordance with good engineering practice and should conform to, but not be limited to, the following standards, codes of practice and specifications:

- Abu Dhabi Municipality (ADM) regulations
- Abu Dhabi Sewerage Services (ADSSC) regulations
- *Uniform Plumbing Code of Abu Dhabi Emirate* (ADEA, 2009)
- American Society of Plumbing Engineers (ASPE) *Plumbing Engineering Design Handbook* (four volumes) (ASPE, 2016–2019)
- British Standards (BS)\* including:
  - BS EN 12056-1 (BSI, 2000a)\*
  - BS EN 12056-2 (BSI, 2000b)\*

- BS EN 12056-4 (BSI, 2000d)\*
- BS EN 12056-5 (BSI, 2000e)\*
- European Norms (EN)\*.

\* Code to be followed only when local and US codes do not provide any specific information.

Note that ADM will approve the building drainage design as per their requirements, and ADSSC will approve the plot connection to the external sewer system as per their requirements, both calculation methodologies have been included in sections 8.1 and 8.2 of this document.

## 8.1 Building system pipe sizing

Building peak black (foul) and grey (waste) water flows should be calculated according to *Uniform Plumbing Code of Abu Dhabi Emirate* (ADEA, 2009) standards. The drainage fixture unit method is the basis of design to calculate peak flow rates and pipe sizes for drainage systems. Note that drainage design in Abu Dhabi is generally based on a system divided into a black water stack serving WCs and urinals and a grey water stack serving all other appliances.

Drainage load in the building drainage system is determined by the total load on the pipe being sized, in terms of drainage fixture units (DFU), as shown in the *Uniform Plumbing Code*, Table 7-3, 'Drainage Fixture Unit Values', reproduced as Table 8.1 below. Once the total DFU has been calculated pipe sizes can be selected utilising *Uniform Plumbing Code* Table 7-4, 'Maximum Unit Loading and Maximum Length of Drainage and Vent Piping', reproduced as Tables 8.2 and 8.3 below.

**Table 8.1** Drainage fixture units (DFUs) (source: *Uniform Plumbing Code of Abu Dhabi Emirate* (ADEA, 2009))

| Appliance, appurtenance, or fixture                                           | Min. trap size and trap arm <sup>[7,8]</sup> (mm) | Drainage fixture unit (DFU) values |                    |                         |
|-------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------|--------------------|-------------------------|
|                                                                               |                                                   | Private                            | Public             | Assembly <sup>[9]</sup> |
| Bidet                                                                         | 32                                                | 1.0                                |                    |                         |
|                                                                               | 40                                                | 2.0                                |                    |                         |
| Clothes washer (domestic), standpipe <sup>[5]</sup>                           | 50                                                | 3.0                                | 3.0                | 3.0                     |
| Dental unit, cuspidor                                                         | 32                                                |                                    | 1.0                | 1.0                     |
| Dishwasher (domestic), with independent drain                                 | 40 <sup>[2]</sup>                                 | 2.0                                | 2.0                | 2.0                     |
| Drinking fountain or watercooler                                              | 32                                                | 0.5                                | 0.5                | 1.0                     |
| Food-waste grinder (commercial)                                               | 50                                                |                                    | 3.0                | 3.0                     |
| Floor drain, emergency                                                        | 50                                                |                                    | 0.0                | 0.0                     |
| Floor drain (for additional sizes see Section 702 of the UPCode (ADEA, 2009)) | 50                                                | 2.0                                | 2.0                | 2.0                     |
| Shower (single-head) trap                                                     | 50                                                | 2.0                                | 2.0                | 2.0                     |
| Shower (multi-head) each additional head                                      | 50                                                | 1.0                                | 1.0                | 1.0                     |
| Lavatory, single                                                              | 32                                                | 1.0                                | 1.0                | 1.0                     |
| Lavatory, in sets of two or three                                             | 40                                                | 2.0                                | 2.0                | 2.0                     |
| Wash fountain                                                                 | 40                                                |                                    | 2.0                | 2.0                     |
|                                                                               | 50                                                |                                    | 3.0                | 3.0                     |
| Mobile home, trap                                                             | 80                                                | 12.0                               |                    |                         |
| Receptor, indirect waste <sup>[1,3]</sup>                                     | 40                                                |                                    | See footnotes 1, 3 |                         |
| Receptor, indirect waste <sup>[1,4]</sup>                                     | 50                                                |                                    | See footnotes 1, 4 |                         |
| Receptor, indirect waste <sup>[1]</sup>                                       | 80                                                |                                    | See footnote 1     |                         |
| Sinks:                                                                        |                                                   |                                    |                    |                         |
| — bar                                                                         | 40                                                | 1.0                                |                    |                         |
| — bar                                                                         | 40 <sup>[2]</sup>                                 |                                    | 2.0                | 2.0                     |
| — clinical                                                                    | 80                                                |                                    | 6.0                | 6.0                     |
| — commercial with food waste                                                  | 402                                               |                                    | 3.0                | 3.0                     |
| — special purpose                                                             | 402                                               | 2.0                                | 3.0                | 3.0                     |
| — special purpose                                                             | 50                                                | 3.0                                | 4.0                | 4.0                     |
| — special purpose                                                             | 80                                                |                                    | 6.0                | 6.0                     |
| — kitchen (domestic), with or without food-waste grinder and/or dishwasher    | 40 <sup>[2]</sup>                                 | 2.0                                | 2.0                |                         |
| — laundry, with or without discharge for a clothes washer                     | 40 <sup>[2]</sup>                                 |                                    | 2.0                | 2.0                     |
| — service or mop basin                                                        | 50                                                |                                    | 3.0                | 3.0                     |
| — service or mop basin                                                        | 80                                                |                                    | 3.0                | 3.0                     |
| — service, flushing rim                                                       | 80                                                |                                    | 6.0                | 6.0                     |
| — wash (each set of faucets)                                                  |                                                   |                                    | 2.0                | 2.0                     |

Table continues

**Table 8.1** Drainage fixture units (DFUs) (*continued*) (source: *Uniform Plumbing Code of Abu Dhabi Emirate* (ADEA, 2009))

| Appliance, appurtenance, or fixture                            | Min. trap size and trap arm <sup>[7,8]</sup> (mm) | Drainage fixture unit (DFU) values |        |                         |
|----------------------------------------------------------------|---------------------------------------------------|------------------------------------|--------|-------------------------|
|                                                                |                                                   | Private                            | Public | Assembly <sup>[9]</sup> |
| — integral trap (2 litre flush)                                | 50 <sup>[2]</sup>                                 | 2.0                                | 2.0    | 5.0                     |
| — integral trap (greater than 2 litre flush)                   | 50                                                | 2.0                                | 2.0    | 6.0                     |
| — exposed trap                                                 | 40 <sup>[2]</sup>                                 | 2.0                                | 2.0    | 5.0                     |
| — non-water supplied                                           | 40 <sup>[2]</sup>                                 | 0.5                                | 0.5    | 0.1                     |
| Water closet:                                                  |                                                   |                                    |        |                         |
| — 6 litre flush, gravity tank <sup>[6]</sup>                   | 80                                                | 3.0                                | 4.0    | 6.0                     |
| — 6 litre flush, flushometer tank <sup>[6]</sup>               | 80                                                | 3.0                                | 4.0    | 6.0                     |
| — 6 litre flush, flushometer valve <sup>[6]</sup>              | 80                                                | 3.0                                | 4.0    | 6.0                     |
| — greater than 6 litre flush, gravity tank <sup>[6]</sup>      | 80                                                | 4.0                                | 6.0    | 8.0                     |
| — greater than 6 litre flush, flushometer valve <sup>[6]</sup> | 80                                                | 4.0                                | 6.0    | 8.0                     |

**Notes:**

- 1 Indirect waste receptor shall be sized based on the total drainage capacity of the fixtures that drain therein to, in accordance with Table 7-2 of the *Uniform Plumbing Code of Abu Dhabi Emirate*.
- 2 Provide a 50 mm (2 in.) minimum drain.
- 3 For refrigerators, coffee urns, water stations, and similar low demands.
- 4 For commercial sinks, dishwashers, and similar moderate or heavy demands.
- 5 Buildings having a clothes-washing area with clothes washer in a battery of three or more clothes washers shall be rated at six fixture unit each for the purposes of sizing common horizontal and vertical drainage piping.
- 6 Water closets shall be computed as six fixtures units when determining septic tank sizes based on Appendix K of the *Uniform Plumbing Code of Abu Dhabi Emirate*.
- 7 Minimum trap size shall be based on the fixture outlet size.
- 8 Trap sizes shall not be increased to the point where the fixture discharge may be inadequate to maintain their self-scouring properties.
- 9 Assembly = public use (see Table 4-1 within the *Uniform Plumbing Code of Abu Dhabi Emirate*).

Unit conversion: 1 mm = 0.04 in.; 1 litre = 0.26 gal.

**Table 8.2** Maximum unit loading units for drainage and vent piping (source: *Uniform Plumbing Code of Abu Dhabi Emirate* (ADEA, 2009))

| Pipe layout                                       | Maximum loading units for stated pipe size (mm) |                  |                   |                   |                   |                    |                    |                     |                     |                     |
|---------------------------------------------------|-------------------------------------------------|------------------|-------------------|-------------------|-------------------|--------------------|--------------------|---------------------|---------------------|---------------------|
|                                                   | 32                                              | 40               | 50                | 65                | 80                | 100                | 150                | 200                 | 250                 | 300                 |
| Drainage piping <sup>[1]</sup> :                  |                                                 |                  |                   |                   |                   |                    |                    |                     |                     |                     |
| — vertical                                        | 1                                               | 2 <sup>[2]</sup> | 16 <sup>[3]</sup> | 32 <sup>[3]</sup> | 48 <sup>[4]</sup> | 256                | 1380               | 3600                | 5600                | 8400                |
| — horizontal                                      | 1                                               | 1                | 8 <sup>[3]</sup>  | 14 <sup>[3]</sup> | 35 <sup>[4]</sup> | 216 <sup>[3]</sup> | 720 <sup>[5]</sup> | 2640 <sup>[3]</sup> | 4680 <sup>[5]</sup> | 8200 <sup>[5]</sup> |
| Vertical stack (exceeding three branch interval): |                                                 |                  |                   |                   |                   |                    |                    |                     |                     |                     |
| — total for 1 branch interval                     | —                                               | 2 <sup>[2]</sup> | 6 <sup>[3]</sup>  | 9 <sup>[3]</sup>  | 20 <sup>[4]</sup> | 90                 | 350                | 600                 | 1000                | 1500                |
| — total for stack                                 | —                                               | 8 <sup>[2]</sup> | 24 <sup>[3]</sup> | 42 <sup>[3]</sup> | 72 <sup>[4]</sup> | 500                | 1900               | 3600                | 5600                | 8400                |
| Vent piping (see note):                           |                                                 |                  |                   |                   |                   |                    |                    |                     |                     |                     |
| — horizontal and vertical                         | 1                                               | 8 <sup>[3]</sup> | 24                | 48                | 84                | 256                | 1380               | 3600                | —                   | —                   |

[1] Excluding trap arm.

[2] Except sinks, urinals, and dishwashers.

[3] Except six-unit traps or water closets.

[4] Only four water closets or six-unit traps allowed on any vertical pipe or stack; not to exceed three water closets or six-unit traps on any horizontal branch or drain.

[5] Based on 20 mm/m (1/4 in./ft.) slope. For 10 mm/m (1/8 in./ft.) slope, multiply horizontal fixture units by a factor of 0.8.

*Note:* The diameter of an individual vent shall be not less than 32 mm (1 1/4 in.) nor less than 1/2 the diameter of the drain to which it is connected. Fixture unit load values shall be not less than 32 mm (1 1/4 in.) nor less than 1/2 diameter of the drain to which it is connected. Fixture unit load values for drainage and vent piping shall be computed from Table 7-3 and section 702.0 of the UPCode (ADEA, 2009). Not to exceed 1/3 of the total permitted length of any vent may be installed in a horizontal position. When vents are increased one pipe size for their entire length, the maximum length limitations specified in this table do not apply.

*Unit conversion:* 1 mm = 3.281 ft.; 1 mm = 0.04 in.

**Table 8.3** Maximum lengths of drainage and vent piping (source: *Uniform Plumbing Code of Abu Dhabi Emirate* (ADEA, 2009))

| Pipe layout               | Maximum length (m) for stated pipe size (mm) |    |    |    |    |     |     |     |
|---------------------------|----------------------------------------------|----|----|----|----|-----|-----|-----|
|                           | 32                                           | 40 | 50 | 65 | 80 | 100 | 150 | 200 |
| Drainage piping:          |                                              |    |    |    |    |     |     |     |
| — vertical                | 14                                           | 20 | 26 | 45 | 65 | 91  | 155 | 229 |
| — horizontal              | Unlimited                                    |    |    |    |    |     |     |     |
| Vent piping (see note):   |                                              |    |    |    |    |     |     |     |
| — horizontal and vertical | 14                                           | 18 | 37 | 55 | 65 | 91  | 155 | 229 |

*Note:* The diameter of an individual vent shall be not less than 32 mm (1¼ in.) nor less than 1/2 the diameter of the drain to which it is connected. Fixture unit load values shall be not less than 32 mm (1¼ in.) nor less than 1/2 the diameter of the drain to which it is connected. Fixture unit load values for drainage and vent piping shall be computed from Table 7-3 and section 702.0. Not to exceed 1/3 of the total permitted length of any vent may be installed in a horizontal position. When vents are increased one pipe size for their entire length, the maximum length limitations specified in this table do not apply.

*Unit conversion:* 1 mm = 3.281 ft.; 1 mm = 0.04 in.

## 8.2 Plot/building system peak flow rate

The figures in Table 8.4 have been taken from Table 3.2.7 in the Abu Dhabi Sewerage Services Company (ADSSC) *Design Guidelines*, Section 3, 'Sewerage System Design' (ADSSC, 2008) to determine the daily sewerage flowrate per capita from the building/plot.

### 8.2.1 Per capita flow

Sewer systems shall be designed on the basis of details given in Table 8.4.

**Table 8.4** Design flows (source: ADSSC (2008))

| Development type        | Occupancy                 | Average daily flow (litres/person)        |
|-------------------------|---------------------------|-------------------------------------------|
| Low-cost residential    | 0–16 persons              | 180 litres                                |
| Medium-cost residential | 0–16 persons              | 225 litres                                |
| High-cost residential   | 0–6 persons               | 275 litres                                |
| Large villas/palaces    | 9–50 persons              | 275 litres                                |
| High-rise               | Number of flats × 5       | 275 litres                                |
| Educational             | Number of pupils + staff  | 70 litres                                 |
| Hospital <sup>[1]</sup> | Number of beds + staff    | 350 litres                                |
| Commercial              | Number of staff/visitors  | 50 litres                                 |
| Mosques                 | Floor area m <sup>2</sup> | 100 litres                                |
| Wet industry            | Not applicable            | Varies to be advised                      |
| Dry industry            | Number of staff           | 50 litres at 8 persons per m <sup>2</sup> |
| Army camps              | Number of occupants       | 100 litres                                |
| Hotels                  | Number of rooms           | 885 litres per room per day               |

[1] Number of persons taken as twice the number of beds.

### 8.2.2 Peak flow

Sewers shall be designed on a peak flow basis using one of the following methods:

- (i) The ratio of peak to average daily flow as determined from the equation:

$$Q_{\max}/Q_{\text{ave}} = (18 + \sqrt{P}) / (4 + \sqrt{P})$$

where  $P$  is the population in thousands.

- (ii) Value established from a study acceptable to ADSSC.
- (iii) Use of other values for peak design flow, if justified on the basis of extensive documentation.

## 8.3 Condensate drainage: design criteria, codes, standards and regulations

The design, specification and installation of condensate drainage systems should be in accordance with good engineering practice and should conform to, but not be limited to, the following standards, codes of practice and specifications:

- *Abu Dhabi International Building Code* (ADDMA, 2014)
- *Uniform Plumbing Code of Abu Dhabi Emirate* (ADEA, 2009).

Condensate from all cooling coils and evaporators shall be conveyed from the drain pan outlet to an approved place of disposal and collected for on-site reuse.

The waste pipe shall have a slope of not less than 10 mm/m ( $1/8$  in./ft.) or 1 percent slope and shall be of approved corrosion-resistant material not smaller than the outlet size as required in Table 8-2 of the *Uniform Plumbing Code of Abu Dhabi Emirate* (ADEA, 2009) for air-cooling coils or condensing fuel-burning appliances, respectively. Condensate or wastewater shall not drain over a public way.

Air conditioning condensate waste pipes shall be independent of any drainage and waste system and shall not be smaller than shown in Table 8-2 of the *Uniform Plumbing Code of Abu Dhabi Emirate* (ADEA, 2009), reproduced below as Table 8.5.

**Table 8.5** Minimum size for condensate pipes

| Equipment capacity (kW) | Condensate pipe minimum diameter (mm) |
|-------------------------|---------------------------------------|
| Up to 70                | 20                                    |
| 71–140                  | 25                                    |
| 141–320                 | 32                                    |
| 321–440                 | 40                                    |
| 441–800                 | 50                                    |

Unit conversion: 1 mm = 0.04 in.

## 9 Stormwater design criteria, codes, standards and regulations

The design, specification and installation of rainwater systems should be in accordance with good engineering practice and should conform with, but not be limited to, the following standards, codes of practice and specifications:

- *Stormwater & Subsoil Drainage Systems: Volume 1: Design Manual* (WA-726-1) (ADDMA, 2016)
- Abu Dhabi Municipality requirements
- *Uniform Plumbing Code of Abu Dhabi Emirate* (ADEA, 2009)
- American Society of Plumbing Engineers (ASPE) *Plumbing Engineering Design Handbook* (four volumes) (ASPE, 2016–2019)
- BS EN 12056-3 (BSI, 2000c)\*
- BS EN 12056-5 (BSI, 2000e)\*
- BS EN 752 (BSI, 2017)\*

\* Code to be followed, only when local and US codes do not provide any specific information)

As per the *Uniform Plumbing Code*, the stormwater system shall be designed based on a return period of 1:100 years with 60 minutes duration.

However, as good design practice taking into consideration climate change, all rainwater downpipes should be sized based on a minimum rainfall intensity of 150 mm per hour i.e. 75 mm per hour for primary drainage and 75 mm per hour for secondary 'emergency overflow' drainage system.

## 10 Fuel gas systems: design criteria, codes, standards and regulations

The design, specification and installation of fuel gas systems should be in accordance with good engineering practice and should conform to, but not be limited to, the following standards, codes of practice and regulations applicable to Abu Dhabi:

- Abu Dhabi Civil Defence requirements
- *UAE Fire and Life Safety Code of Practice* (ADCD, 2018)
- NFPA 54: *National Fuel Gas Code* (NFPA, 2018) (if LPG is utilised)
- NFPA 58: *Liquefied Petroleum Gas Code* (NFPA, 2020) (if LPG is utilised)
- Institute of Gas Engineers and Managers (IGEM) Technical Standards (<https://www.igem.org.uk>).

## 11 Water services: design considerations

### 11.1 Cold water storage, treatment and distribution

The cold water supply design must take into account the high summer temperatures of mains cold water in Abu Dhabi (peaking at approximately 46 °C at the height of summer). For some applications, such as commercial five-star hotels, hospitals, schools and sports complexes, the storage tank will need to be split to allow for a cooled cold water section. This will allow the water to be chilled to a temperature below 20 °C via plate heat exchangers served by

the centralised chiller plant. The cold water supply in raw (ambient) water tanks should be filtered and treated before being stored and cooled (where required) for distribution to the building.

Additional bacterial protection should be incorporated to protect against the risk of *Legionella* as described under section 6.

## 11.2 Hot water generation and distribution

The climate conditions year-round in Abu Dhabi are such that there is no need for heating systems within buildings, thus no heating boilers are required. The primary hot water requirements for most buildings are for domestic use and for applications whereby significant volumes of hot water are required. Domestic hot water generation shall primarily be from solar energy based on a sustainability point of view.

In instances whereby relatively low volumes are required, or in remote areas of a building or development, the use of localised hot water heating should be considered.

The type of system selected to generate the heat input to the solar thermal system should be based on high-performance solar collectors. The final systems selected will also need to be appropriate for Abu Dhabi climatic conditions. Solar hot water systems should also be selected according to project size: a thermosiphon system is the better option for small villas or projects, whereas a drain-back system would be the better option for medium to large scale projects.

Solar hot water systems should be designed to achieve maximum efficiency. The energy back-up for the system can be water-to-water heat pumps, electric coil, fuel-fired boiler or any other suitable system.

Solar hot water systems may be considered for achieving Estidama credits for Renewable Energies.

Hot water pasteurisation should be provided as described in section 6.

## 12 Foul and wastewater drainage: system design considerations

### 12.1 Underground drainage

There should be separate underground networks for the ground floor and upper floors. A stormwater disposal system should be provided to convey the discharge from above-ground sanitation and within the building and, if applicable, any external car parking area. Foul and stormwater drainage systems should be designed to meet the requirements of *Stormwater & Subsoil Drainage Systems: Volume 1: Design Manual* (WA-726-1) (ADDMA, 2016), Abu Dhabi Sewerage Services and Abu Dhabi Municipality regulations.

### 12.2 Above-ground drainage

A foul water drainage disposal system should be provided to serve the domestic sanitary fittings within new developments. The design of the system should be in accordance with *Uniform Plumbing Code of Abu Dhabi Emirate* (ADEA, 2009) guidance and Abu Dhabi municipality requirements.

Foul and waste water drainage systems should be designed to safely carry away discharges from sanitary appliances in a quick and quiet manner without causing a nuisance or risk to health.

Pipework, fittings and accessories should be installed to ensure that:

- generally, all foul and wastewater discharge is conveyed under gravity conditions without crossflow or back-fall to prevent leakage and/or blockages occurring
- foul air from the drainage system does not enter the building
- pressure fluctuations within the pipework system do not vary by more than 50 mm water gauge
- all pipework located within main risers, shop units and plant room areas is fully accessible
- rodding access is provided at all changes of direction, branch connections and at each floor level
- pipework systems can be adequately tested, cleaned and maintained
- pipework to be run within ceiling voids is kept to a minimum and installed to the shortest practical route
- all pipework passing through adjoining fire compartments is fire-sleeved and sealed.

For high-rise buildings, any vertical discharge, waste, ventilating and rainwater pipes over six storeys in height will be a minimum of 150 mm diameter. Ground floor shall be connected separately to the below-ground drainage and velocity breaks installed. This is to avoid problems with backpressures pushing out traps and for compliance with Abu Dhabi Municipality regulations.

The local foul water infrastructure should be investigated to determine final connection points, agreed permissible flows and invert levels into the local authority network.

Under no circumstances should a foul water soakaway be incorporated into the design. Foul water must not be allowed to soak into the ground, as this will have an impact on groundwater conditions and could lead to public health risks to the surrounding area. Septic-type tanks should be utilised in circumstances where there is no local connectivity to the municipality sewer system. These would ideally be located external to the building in an accessible location, and need to be emptied on a regular basis by an authorised waste management company.

The foul and waste water drainage system should be designed to safely carry water from sanitary appliances via gravity discharges in a quick and quiet manner without causing a nuisance or risk to health.

The systems should comprise:

- external trapped gullies
- grease interceptors
- petrol interceptors
- lint interceptor
- sand interceptor
- oil interceptor
- neutralisation system
- interconnecting gravity drainage pipework
- manholes
- rodding access points
- linear drainage channels.

## 12.3 Kitchen drainage

Article 19 of the Abu Dhabi Food Control Authority's Regulation No (6) of 2010: *Food Hygiene Throughout the Food Chain* (ADFCA, 2010) requires that:

'Drainage facilities shall conform to the following conditions:

- a. Shall be of adequate size and design and appropriately installed and maintained for the intended purposes to avoid the risk of contamination and foul odors.
- b. Drainage channels, including any removable perforated baskets and anti-slip grating, shall be so constructed from suitable materials and be kept in such good order, repair and condition as to minimise any risk of contamination.
- c. Where drainage channels are fully or partially open, they shall be designed so as to ensure that waste does not flow from a contaminated area towards or into a clean area, in particular an area where high risk food is handled.'

Guidance on compliance with ADFCA legislation is provided in Code of Practice No. (13)/2011: *Food Hygiene for Food Service* (ADFCA, 2011).

Adequate drainage and waste disposal systems and facilities should be provided in all food businesses. They should be designed and constructed so that the risk of contaminating food or the potable water supply is avoided. To be effective, all sewage and waste water must be conveyed from all buildings on the site so as not to cause ponding or backflow into the building. Sewage and wastewater disposed of on-site must be disposed of so as not to jeopardise food safety.

The disposal system must also be constructed so that there is no likelihood of the liquid waste polluting the water supply or contaminating food. This is intended to ensure that the standard of workmanship is such that the system is not likely to leak, block, overflow or allow access by vermin into the food premises.

## 12.4 Interceptors

### 12.4.1 Grease interceptors

The foul drainage system from commercial kitchens or retail food and beverage areas should be fully protected with the use of suitably-sized grease removal units. These should be located in areas that are accessible for regular cleaning and maintenance, and should be externally located where possible. Grease separators should be designed as per BS EN 1825-1 (BSI, 2004).

### 12.4.2 Oil interceptors

Generator rooms, plumbing workshop areas and oily waste areas should be discharged into oil interceptors before connecting to the drainage network.

### 12.4.3 Lint interceptors

Lint interceptors should be provided for commercial laundry prior to discharge to the drainage network.

### 12.4.4 Laboratory waste neutralisation system

Laboratory waste should be discharged through a neutralisation pit before connecting to the soil network, and a separate vent pipe should be provided for the laboratory system.

### 12.4.5 Petrol interceptors

Suitably-sized petrol interceptors should be provided for discharges from all car parking areas, and should be located in areas that are accessible for regular cleaning and maintenance. These should be provided with double-sealed recessed covers suitably rated for the expected traffic flow over manholes.

## 13 Stormwater design considerations

A separate below-ground stormwater drainage system should be provided to convey all rainwater discharge from roofs and open-ground hard-paved areas, via gravity, to the Abu Dhabi Municipality connection manhole located on the periphery of the building plot. The local surface water infrastructure should be further investigated to determine final connection points, agreed permissible flows and invert levels into the local Municipality authority network.

Flows from roofs and paved areas should be based on based on a 1:100-year storm period with a duration of 60 minutes. The rainfall intensity for return period shall be taken from Table 3-4 in *Stormwater & Subsoil Drainage Systems: Volume 1: Design Manual* (WA-726-1) (ADDMA, 2016), reproduced below as Table 13.1.

**Table 13.1** Rainfall intensity duration frequency (source: ADDMA, 2016)

|                       | Duration (minutes)           |        |        |       |       |       |       |       |      |      |
|-----------------------|------------------------------|--------|--------|-------|-------|-------|-------|-------|------|------|
|                       | 5                            | 10     | 15     | 30    | 60    | 120   | 180   | 360   | 720  | 1440 |
|                       | Duration (hours)             |        |        |       |       |       |       |       |      |      |
|                       | 0.083                        | 0.167  | 0.25   | 0.5   | 1     | 2     | 3     | 6     | 12   | 24   |
| Return period (years) | Rainfall intensity (mm/hour) |        |        |       |       |       |       |       |      |      |
| 100                   | 277.89                       | 172.13 | 130.07 | 80.57 | 49.91 | 30.91 | 23.36 | 14.47 | 8.96 | 5.55 |
| 50                    | 220.26                       | 136.43 | 103.09 | 63.86 | 39.56 | 24.50 | 18.51 | 11.47 | 7.10 | 4.40 |
| 25                    | 176.1                        | 108.7  | 81.97  | 50.6  | 31.24 | 19.28 | 14.54 | 8.98  | 5.54 | 3.42 |
| 15                    | 150.82                       | 92.97  | 70.05  | 43.18 | 26.62 | 16.41 | 12.36 | 7.62  | 4.70 | 2.90 |
| 10                    | 123.61                       | 76.09  | 57.29  | 35.26 | 21.71 | 13.36 | 10.06 | 6.19  | 3.81 | 2.35 |
| 5                     | 91.71                        | 56.03  | 41.99  | 25.65 | 15.67 | 9.57  | 7.18  | 4.38  | 2.68 | 1.64 |
| 2                     | 51.89                        | 31.07  | 23.02  | 13.78 | 8.25  | 4.94  | 3.66  | 2.19  | 1.31 | 0.79 |

Any roof area should have at least two rainwater outlets. For large roof areas, a secondary means of overflow must be implemented.

The roofs and hard-standing areas of all buildings should be connected to the stormwater network comprising roof outlets and, where appropriate, floor gullies. Surface water should be drained from roof areas via fixed downpipes connecting into the below-ground drainage system. Downpipes should be integrated into the building fabric or mounted externally.

The rainwater drainage system should be designed to operate under gravity conditions, and should be designed to ensure that water is removed from the roof as quickly and effectively as possible to prevent any water ingress into the building. This system consists of roof outlets draining into rainwater downpipes before exiting the building to connect to the drainage infrastructure manhole.

Access points should be provided throughout the rainwater drainage system at all changes of direction and at each floor level to facilitate future maintenance.

All external hard standing areas should be drained via drainage channel and/ or floor gullies, as agreed with the landscape architect, and should connect to the site-wide network. External areas should not be permitted to flood during storm conditions, drainage must be in place to deal with this event.

### 13.1 Soakaway design

Should a connection not be local to the site or development, the use of suitably-sized soakaways is permissible. These should be designed in line with DG 365: *Soakaway design* (BRE, 2016).

The design method for sizing the soakaway should be based on the equation of volumes:

$$I - O = S$$

where  $I$  is the inflow from the impermeable area drained to the soakaway ( $\text{m}^3$ ),  $O$  is the outflow infiltrating into the soil during rainfall ( $\text{m}^3$ ) and  $S$  is the required storage in the soakaway to balance the temporary inflow and outflow ( $\text{m}^3$ ).

Inflow to the soakaway:

$$I = A \times R$$

where  $A$  is the impermeable area drained to the soakaway ( $\text{m}^2$ ) and  $R$  is the total rainfall in a design storm ( $\text{m}$ ).

Outflow to the soakaway:

$$O = a_{s50} \times f \times D$$

where  $a_{s50}$  is the internal surface area of the soakaway to 50% effective depth (excluding the base area, which is assumed to clog with fine particles and become ineffective in the long term) ( $\text{m}^2$ ),  $f$  is the soil infiltration rate determined by use of a trial pit at the site of the soakaway ( $\text{m/s}$ ) and  $D$  is the storm duration.

Stormwater attenuation should be based on a 1:5-year storm event with a duration of 24 hours, as per local authority guidelines.

It is very important to note that surface water soakaways should not normally be constructed closer than five metres from building foundations. Site investigations must be undertaken to determine the soil conditions, as this is crucial in the viability and sizing of the soakaway.

Flooding is a yearly recurring event in Abu Dhabi and the main reason is the selection of storm event, and the municipality storm water network being overloaded. It is encouraged to use sustainable drainage systems (SuDS) such as infiltration, rain gardens, green roofs, blue roofs etc., to minimise the risk of flooding on a site.

## 14 Fuel gas design considerations

The design of the natural gas (NG), synthetic natural gas (SNG) and liquefied petroleum gas (LPG) systems should prioritise the safety of human beings, whether or not they are installing the system. Modern technology should be adopted to detect any failure in the system through a high-tech gas control panel, linked to the fire control panel, in order to ensure the maintenance of a safe environment.

Design of liquid gas distribution systems should be in accordance with certified and applicable specifications and engineering standards and must be submitted with the drawings for approval of building permits transactions.

Only materials, controls and reading devices approved by Abu Dhabi Civil Defence should be used.

Where on-site storage of SNG and/or LPG is utilised it is preferable to place the LPG tank within the limits of the plot, whether buried, mounded, ground level or above, and with safe distances compatible with national safety requirements and the requirements of Abu Dhabi Civil Defence. Note that SNG distribution systems are preferred over LPG distribution and therefore the LPG is to be mixed with compressed air to ensure SNG is created for distribution.

Note that for the Abu Dhabi Emirate the Civil Defense Authority does not recommend gas installations within basement areas.

The central tank should be supplied with means of security, protection and control in accordance with standards specified by Abu Dhabi Civil Defence, and should comply with all standards and special requirements for public safety adopted by Abu Dhabi Municipality.

Execution of domestic gas networks should be performed only by specialised companies registered with the Abu Dhabi Municipality.

First stage (with two safety devices):

- natural gas and synthetic natural gas entry to building: 75 mbar
- LPG entry to building: 350 mbar (as per NFPA 58, Section 3.2.13 C, (NFPA, 2020), this can be up to 1.4 bar).

Second stage (with one safety device):

- natural gas and synthetic natural gas entry to building: 21 mbar
- LPG inside kitchen: 37 mbar.

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